

Design a transmission line segment that matches $300\ \Omega$ to $50\ \Omega$ at 10 GHz using a quarter-wave match. Assume microstrip line for which propagation occurs with wavelength 60% that of free space.

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The free-space wavelength $\lambda_0 = c/f$ at 10 GHz is $\cong 3\text{ cm}$

the wavelength of the signal in the line $\lambda_g = 0.6\lambda_0 \cong 1.8\text{ cm}$

the length of the line should be $l = \lambda_g/4 \cong 4.5\text{ mm}$.

$$Z_0 = \sqrt{300\ \Omega \cdot 50\ \Omega} \cong 122.5\ \Omega$$

